The C# Language-Integrated Query (LINQ)

🔍 What is LINQ?

LINQ (Language-Integrated Query) is a feature in C# that allows you to query data (arrays, lists, databases, XML, etc.) using a **SQL-like syntax directly inside C#**.

👉 It helps you:

- Filter data
- Sort data
- Group data
- Transform data

🧠 Basic LINQ Concepts

1. Query Syntax (SQL-like)

```
var result = from n in numbers
             where n > 5
             select n;
```

2. Method Syntax (Lambda Expressions)

```
var result = numbers.Where(n => n > 5);
```

📊 Example: Query an Array using LINQ

```
using System;
using System.Linq;

class Program
{
   static void Main()
   {
      int[] numbers = { 1, 3, 5, 7, 9, 2, 4, 6, 8 };

      // Query syntax
      var evenNumbers = from n in numbers
                        where n % 2 == 0
                        orderby n
                        select n;

      Console.WriteLine("Even Numbers:");
      foreach (var num in evenNumbers)
      {
         Console.Write(num + " ");
      }
   }
}
```

✅ Output:

```
Even Numbers:
2 4 6 8
```

⚙️ Common LINQ Methods

Method	Description	Example
Where()	Filters data	numbers.Where(x => x > 5)
Select()	Transforms data	numbers.Select(x => x * 2)
OrderBy()	Sort ascending	numbers.OrderBy(x => x)
OrderByDescending()	Sort descending	numbers.OrderByDescending(x => x)
First()	Gets first element	numbers.First()
Count()	Counts items	numbers.Count()

File Handling in C#

📂 What is File Handling?

File handling allows your program to:

- Create files
- Read files
- Write data
- Update existing files

🧱 8.1 Data Hierarchy

Bit → Byte → Field → Record → File → Database

📁 8.2 Files and Streams

- **File** = storage (e.g., .txt, .csv)
- **Stream** = flow of data between program and file

📦 8.3 File and Directory Classes

🔹 File Class Example

```
using System.IO;

File.WriteAllText("test.txt", "Hello World!");
string content = File.ReadAllText("test.txt");

Console.WriteLine(content);
```

🔹 **Directory Class Example**

```
using System.IO;

Directory.CreateDirectory("MyFolder");

string[] files = Directory.GetFiles(".");

foreach (var file in files)
{
   Console.WriteLine(file);
}
```

📝 **8.4 Text File Handling**

🔹 **Using StreamWriter (Write File)**

```
using System.IO;

StreamWriter writer = new StreamWriter("data.txt");
writer.WriteLine("Hello, this is a file!");
writer.Close();
```

🔹 **Using StreamReader (Read File)**

```
using System.IO;

StreamReader reader = new StreamReader("data.txt");
string text = reader.ReadToEnd();
reader.Close();

Console.WriteLine(text);
```

🔄 Using FileStream (Advanced Control)

```
using System;
using System.IO;

FileStream fs = new FileStream("data.txt", FileMode.OpenOrCreate);

byte[] data = System.Text.Encoding.UTF8.GetBytes("Hello FileStream");
fs.Write(data, 0, data.Length);

fs.Close();
```

🔍 Using LINQ with Files (Search Directories)

```
using System;
using System.IO;
using System.Linq;

class Program
{
   static void Main()
   {
      var files = Directory.GetFiles(".", "*.txt");

      var largeFiles = files
         .Select(f => new FileInfo(f))
         .Where(f => f.Length > 100);

      foreach (var file in largeFiles)
      {
         Console.WriteLine(file.Name);
```

```
    }
  }
}
```